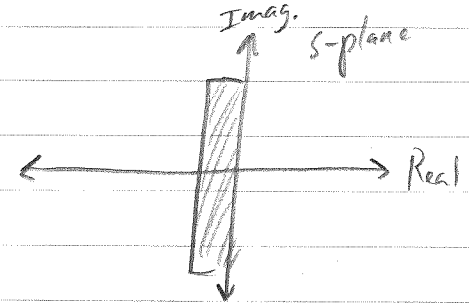
Benjamin Perkins Homework 6 Digital Signal Processing

6.1 Draw a rough sketch of the Laplace s-plane showing a shaded area on the s-plane that corresponds to the shaded circular band in Figure P6-1



6.2 Write the $H(z)$ z-domain transfer equations for the filters describe by the following difference equations

(a) $y(n) = x(n) - y(n-2)$

$$y(n) + y(n-2) = x(n)$$

$$H(z) = \frac{1}{1+z^{-2}}$$

(b) $y(n) = x(n) + 3x(n-1) + 2x(n-2) - y(n-3)$

By inspection, $H(z) = \dfrac{1+3z^{-1}+2z^{-2}}{1+z^{-3}}$

(c) $y(n) = x(n) + x(n-1) + x(n-3) + x(n-4) - y(n-2)$

By inspection, $H(z) = \dfrac{1+z^{-1}+z^{-3}+z^{-4}}{1+z^{-2}}$

6.3 State the order for the filters in P 6.2

a) 2
b) 3
c) 4

6.6 $H(z) = \dfrac{z^2+0.3z+1}{z^2+0.3z+0.8}$

a) What is the transfer function equivalent to $H(z)$ with z having negative exponents

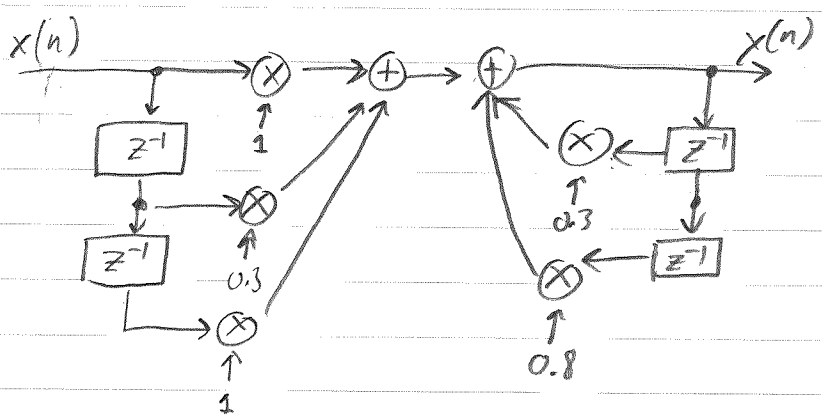
$$H(z) = \frac{1+0.3z^{-1}+z^{-2}}{1+0.3z^{-1}+0.8z^{-2}}$$

b) Is this IIR filter stable?
Yes. The roots of the denominator are

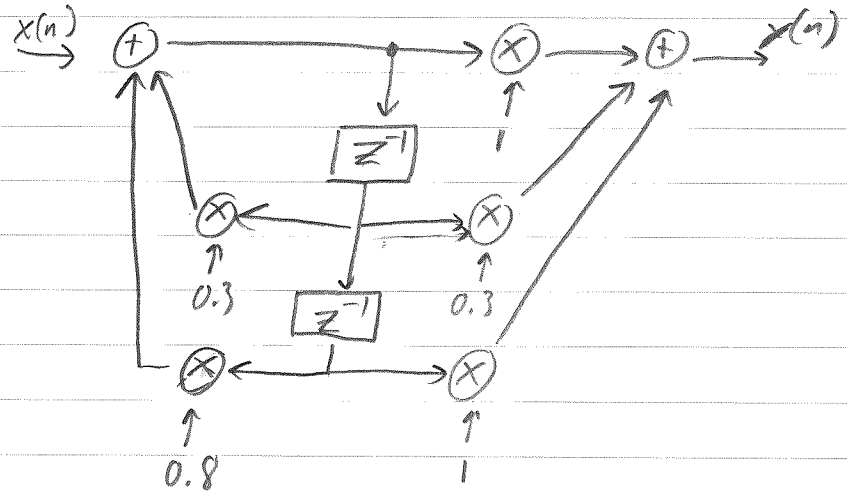
$$z = -0.15 \pm j0.8818$$

Thus, the poles are inside the unit circle.

c) Draw the Direct Form I structure



d) Draw the Direct Form II structure



6.10. Consider the two filters in Figure P6-10. Determine whether each filter is an IIR or an FIR filter.

Both are IIR filters since they both have feedback.